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AI account: Gene Lowelle Roallos – genelowelle06@gmail.com

Full prompt:

I am an engineering student and i need to do a HTML with data visualization. I need you to do some calculation, derivatives, integral, and some stat and some curve fitting. I need a python script that will generate the html file. for now dont do anything and wait for my instruction.

Understood. I'm ready and waiting for your specific instructions on the calculations, data, visualizations, or any other details for the Python script that will generate the HTML file. Just let me know when you're ready.

Next ill show you the file location and dont do anything yet.

Understood. Waiting for the file location and further instructions.

"C:\Users\lowelle\Downloads\Data 01.xlsx" this is the location of the file, dont do anything yet

Understood. File noted: `C:\Users\lowelle\Downloads\Data 01.xlsx`.

Waiting for your next instructions.

i'll give you the python script last week since this is the continuation of this week's activity. I need you to create 3 tabs for this activity. do the Tab 1 first "The derivative you already have Carry the previous meeting's work forward rather than starting again. Recompute the finite-difference derivatives from the raw log, using forward and backward differences at the two ends and central differences in between, and keep the second derivative as well. Convert the timestamps to a numeric axis before anything else. Hours elapsed from the first reading is the sensible choice, and 0.25 h per step keeps the arithmetic honest. Report the time of maximum dh/dt and its value in meters per hour. Say in one sentence what the second derivative told you about the inflow, not just where it changed sign. Keep the same time origin and the same units for the rest of the activity. Half the errors in this laboratory come from mixing minutes, hours and sample indices on the same axis." then do the Tab 2 "The fit, and the arithmetic behind it CHOOSE A MODEL, AND DEFEND IT Levenberg-Marquardt fits the model you give it. It cannot tell you that the model is wrong, only that it cannot get any closer. State your choice in two sentences before you fit anything. CANDIDATE FORM WHAT IT ASSUMES ABOUT THE RESERVOIR Four-parameter logistic h = c + a / (1 + exp(-k (t t0))) One filling event, a single inflection, level settling toward a ceiling. Gompertz h = c + a exp(-exp(-k (t t0))) The same, but rising sharply and easing off slowly. Asymmetric. Sum of two logistics h = c + a1 S1(t) + a2 S2(t) Two pulses of inflow, for example a second rainfall band. Cubic polynomial h = a t3 + b t2 + c t + d Nothing physical. Useful only as a baseline to beat. RUN THE FIT # t in hours from the first reading, h in meters popt, pcov = curve_fit(model, t, h, p0=p0, method="lm", maxfev=20000) Levenberg-Marquardt does not accept bounds. Everything you would have expressed as a bound has to live in p0 instead, so state your initial guess and where each number in it came from. Reading the ceiling and the inflection off the raw plot is a legitimate source. Leaving p0 at its default of ones is not. COMPUTE THESE, IN PYTHON, IN THIS ORDER QUANTITY DEFINITION I N PYTHON Fitted parameters b1 ... bp popt Residuals e i = hi − ĥ(t i) resid = h - model(t, *popt) Prepared by Engr. Mark Airol Escranda · Issued August 25, 2026 2 / 6 DEFINITION I N PYTHON NM-LAB-08252026 NUMERICAL METHODS · LABORATORY ACTIVITY 02 QUANTITY Sum of squared errors SSE = sum of e i 2 sse = np.sum(resid**2) Total sum of squares SST = sum of (hi − mean h)2 sst = np.sum((h - h.mean())**2) Coefficient of determination R2 = 1 − SSE / SST r2 = 1 - sse/sst Standard error of the estimate s = sqrt[SSE / (n − p)] s = np.sqrt(sse/(n - p)) Standard error of a parameter SE(b j) = sqrt[diag(pcov) j] se = np.sqrt(np.diag(pcov)) t statistic t j = b j / SE(b j) tj = popt/se P-value, two tailed p j = 2 [1 − Ft (|t j |, n − p)] 2*(1 - stats.t.cdf(abs(tj), n-p)) n is the number of readings, p the number of fitted parameters, so the degrees of freedom are n − p. Report each parameter with its standard error, its t statistic and its p-value. The logger reads to the centimeter, so a parameter quoted to ten decimal places is a lie about your own data. READ THE RESIDUALS, DO NOT JUST PLOT THEM Plot e i against time and against the fitted value ĥ, then answer in writing: Is the scatter centred on zero throughout, or does it drift? Are there long runs of the same sign? Runs mean the shape of your model is wrong. They do not mean the data is noisy. Does the spread widen as the level rises? Is the largest residual bigger than the logger's own resolution of one centimeter, and by how much? A high R2 sitting on top of patterned residuals is a failed fit. If that is what you find, say so and keep the evidence. An honest failure is worth more marks here than a flattering number." and the Tab 3 "The area under the fitted level Integrate the fitted function, with the x-axis as the reference line and the first and last logged times as the limits: A = ∫t n t 0 ĥ(t) dt Integrate the fitted function with scipy.integrate.quad, not the raw points. Report the value together with the absolute error estimate that quad returns. State the units. If h is in meters and t in hours, the area is in meter-hours, and it is not a volume. Cross-check with np.trapezoid on the raw readings. The two will not agree exactly. Explain the gap in one sentence. Say what the number means to the flood-control office, and what it does not mean." generate the python script now

The Python script has been generated and saved as:

`/home/workdir/artifacts/lab02_depth_analysis.py`

What the script does

1. **Loads** your Excel file (`C:\Users\lowelle\Downloads\Data 01.xlsx`) using the same robust loader from last week's activity.

2. **Tab 1 – Derivatives**
  - Converts timestamps to **hours elapsed from the first reading**.
  - Recomputes 1st and 2nd finite-difference derivatives (forward at the start, backward at the end, central in the interior).
  - Units are **m/h** and **m/h^2**.
  - Reports the exact time and value of maximum `dh/dt`.
  - Gives a one-sentence physical interpretation of the second derivative regarding inflow.

3. **Tab 2 – Curve fit & residuals**

  - Explicitly defends the **four-parameter logistic** model *before* any fitting.
  - Supplies a data-derived initial guess `p0` (not the default ones).
  - Runs Levenberg-Marquardt (`curve_fit(..., method="lm")`).
  - Computes, in the exact order required: parameters, residuals, SSE, SST, R^2, standard error of the estimate, SE of each parameter, t-statistics, and two-tailed p-values.
  - Produces residual-vs-time and residual-vs-fitted plots plus written answers to all the residual-diagnostic questions.

4. **Tab 3 – Integration**
  - Integrates the fitted function with `scipy.integrate.quad`.
  - Reports the value + absolute error estimate (units: **m·h**).
  - Cross-checks with `np.trapezoid` on the raw points and explains the gap in one sentence.
  - States what the number means (and does not mean) for the flood-control office.

5. **Output**
  - A single self-contained HTML file (`Lab02_Reservoir_Stage_Report.html`) with three Bootstrap tabs and interactive Plotly charts.

How to run it

On your machine (with the Excel file in the stated location):

```bash
python lab02_depth_analysis.py
```

Then open `Lab02_Reservoir_Stage_Report.html` in a browser.

Let me know if you want any changes (different model, extra plots, different wording, etc.).

VSC Agent: Copilot

Full Prompts:

fix error

fix the html to look something like this

the graphs are blanks

Change name into Roallos, Gene Lowelle

change colors

can you improve the graph in stage log, Tab 1 (raw log, first derivative, second derivative), Tab 2 (observed stage and four parameter log fit), and Tab 3 (Area under the fitted curve)

can you compile all the prompt here into a pdf file. and save it to my desktop

cancel the pdt command

rename test1.py to lab02_roallos.py

can you make me a pdf file named lab02_roallos_results.pdf. containing one page. Fitted parameters with standard errors, t and p, SSE, R2, s, the area with its units, and your reading of the residuals. save it in my desktop

fix the error in lab02_roallos_results.pdf